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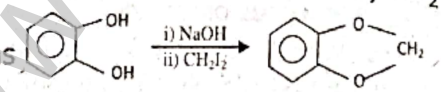
Virudhunagar District
Common Quarterly Examination - 2025

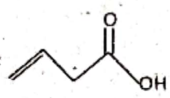
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Standard 12
CHEMISTRY
PART - I

Time: 3.00 Hours | Marks: 70

Answer all the questions **15x1=15**

1) In the extraction of aluminium from alumina by electrolysis, cryolite is added to
   a) Lower the melting point of alumina b) Remove impurities from alumina
   c) Decrease the electrical conductivity d) Increase the rate of reduction
2) The geometry at which carbon atom in diamond are bonded to each other is
   a) Tetrahedral b) Hexagonal c) Octahedral d) None of these
3) Solid (A) reacts with strong aqueous NaOH liberating a foul smelling gas (B) Which spontaneously burn in air giving smoky rings. A and B are respectively
   a) P_4(red) and PH_3 b) P_4(white) and PH_3 c) S_8 and H_2S d) P_4(white) and H_2S
4) Which of the following is weakest acid among all?
   a) HI b) HF c) HBr d) HCl
5) Which of the following lanthanoid ions is diamagnetic?
   a) Eu^{2+} b) Yb^{2+} c) Ce^{2+} d) Sm^{2+}
6) Potassium has a bcc structure with nearest neighbour distance 4.52 $\overset{o}{A}$. Its atomic weight is 39, its density will be
   a) 915 kgm^{-3} b) 2142 kgm^{-3} c) 452 kgm^{-3} d) 390 kgm^{-3}
7) The addition of a catalyst during a chemical reaction alters which of the following quantities?
   a) Enthalpy b) Activation energy c) Entropy d) Internal energy
8) What is the order of the Isomerisation of propene
   a) one b) zero c) three d) two
9) The aqueous solutions of sodium formate, anilinium chloride and Potassium cyanide are respectively
   a) acidic, acidic, basic b) basic, acidic, basic
   c) basic, neutral, basic d) none of these
10) What is the value of Inoic product of water at room temperature?
   a) 1×10^{-7} b) 1×10^{-14} c) 1×10^{-10} d) none of these
11) Calculate the pH of 0.001 M HCl solution
   a) 10^{-3} b) −3 c) 3 d) 3.3
12) Which one of the following will react with phenol to give salicylaldehyde after hydrolysis
   a) Dichloro methane b) trichloromethane
   c) trichloro methane d) CO_2
13) The reactions 

   a) Wurtz reaction b) cyclic reaction
   c) williamson reaction d) Kolbe reaction
14) Reaction of acetone with one of the following reagents involves nucleophilic addition followed by elimination of water. The reagent is
   a) Grignard reagent b) Sn/HCl
   c) Hydrazine in presence of slightly acidic solution d) Hydrocyanic acid
15) The IUPAC name of 

   a) but-3-enoic acid b) but-1-1-ene-4-oicacid
   c) but-2-ene-1-oic acid d) but-3-ene-1-oicacid

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PART - II

Answer 6 six questions. Q.No. 24 is compulsory. **6x2=12**

16) Give two limitations of Ellingham diagram
17) How will you convert boric acid to boron nitride?
18) What happens when PCl_5 is heated?
19) Why do Zirconium and Hafnium exhibit similar properties?
20) Why ionic crystals are hard and brittle?
21) Define pH
22) How will you convert cumene to phenol?
23) Explain Benzoin condensation reaction
24) The rate constant for a first order reaction is $1.54 \times 10^{-3} s^{-1}$. Calculate its half life period

PART - III

Answer 6 six questions. Q.No. 33 is compulsory. **6x3=18**

25) Explain the principle of electrolytic refining with an example
26) Preparation of Borazole (or) Borazine from diborane
27) What is the hybridisation of iodine in IF_7? Give its structure
28) What is Hume - Rothery rule about alloy formation of transition metals?
29) What are point defects? Give the classification flow chart.
30) Give the differences between rate of a reaction and rate constant of a reaction
31) Explain common ion effect with an example
32) Preparation of Picric acid from phenol
33) What is Urotropine? Give the preparation, structure and uses

PART - IV

Answer all the questions. **5x5=25**

34) a) Explain Froth Flotation process with example and diagram

(OR)

b) i) Preparation of Aluminium chloride by McAfee Process
ii) Give the uses of silicones

35) a) What is lanthanoid contraction and what are the effects of lanthanoid contraction?

(OR)

b) i) What is inert pair effect?
ii) Deduce the oxidation number of oxygen in hypofluorous acid - HOF

36) a) Derive integrated rate low for a first order reaction

(OR)

b) i) Explain briefly seven types of unit cell
ii) Write a note on covalent solid with one example

37) a) Derive an expression for the hydrolysin constant and degree of hydrolysis of salt of strong acid and weak base

(OR)

b) i) What are Lewis acids and bases? Give two examples for each
ii) What is Packing fraction (or) Packing efficiency?

38) a) Write the mechanism of Aldol condensation reaction.

(OR)

b) Write short notes on the following
i) Swern oxidation reaction
ii) Schotten - Baumann reaction
iii) Rosenmund reduction reaction

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